

Model scraping for comparison and validation

Summary						
Model name:	Ceftaz_1	Population type*:		ICU		
Drug*:	Ceftazidime	Model type:		POPPK		
Administration route/mode:	discount. IV (17%)/continuous (83%)	No. compartments*:		1		
Publication						
Title:	Population Pharmacokinetics and Probability of Target Attainment of Different Dosing Regimens of Ceftazidime in Critically Ill Patients with a Proven or Suspected Pseudomonas aeruginosa Infection					
Author(s):	Annabel Werumeus Buning, Caspar J Hodiament , Natalia M Lechner , Margriet Schokkin , Paul W G Elbers, Nicole P Juffermans, Ron A A Mathôt, Menno D de Jong , Reinier M van Hest					
Other info:	PMID 34063815					
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/34063815/					
Source Study & Dataset						
Source study:	NA					
Inclusion criteria:	ICU Patients aged ≥18 years treated with IV ceftazidime for a proven or suspected clinically relevant <i>P. aeruginosa</i> infection At least one detectable ceftazidime serum concentration available during the course of therapy Excluded : Cystic fibrosis patients					
Patient disease(s):	See table					
Drug dosing units*:	mg	Number of patients:		96		
Plasma conc. units*:	mg/L	No. samples:		394		
Daily dosing scheme	See table					
Notes/further info:	Continuous infusion : 83% of patients					
Patient Characteristics						
Continuous covariate	Units	Median	Mean	SD	Mode	Range
Age	Y	59				20-84
Body weight	Kg	79				44-237
Body mass index	kg/m²	25				16-66
SOFA score at start of ceftazidime therapy		10				4-16
Creatinine	mg/dL	0.98				0.19-7.49
eGFR*	mL/min/m²	73				6-153
Categorical covariate	Feature			Count (%)		
Sex Male	Yes			60		
Daily dose <3 g continuous infusion	Yes			1		
Daily dose 3 g continuous infusion	Yes			35		
3–5 g continuous infusion	Yes			11		
Daily dose 5 g continuous infusion	Yes			25		
Daily dose 6 g continuous infusion	Yes			9		
Loading dose	Yes			81		
30-days mortality	Yes			39		

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Primary infection site	Pneumonia	39
Primary infection site	Abdominal	14
Primary infection site	Bloodstream	18
Primary infection site	Meningitis	23
Admission category	Surgical	44
Vasopression	Yes	69
Ventilation	Yes	77
Minimum Inhibitory concentration (mg/L) <i>P. aeruginosa</i> at start therapy	2	40
Minimum Inhibitory concentration (mg/L) <i>P. aeruginosa</i> at start therapy	4	25
CVVH	Yes	21
Notes/further info:	* using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formula	

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1 st order):	1 st		
Estimation algorithm:	NA	Log-transformed?	Yes
BLLOQ handling method (mg/L)	0.5	Dose compartment:	Central
All covariates tested:	Age sex weight BMI serum sodium norepinephrine use (yes/no) furosemide use (yes/no) CVVH (yes/no) CLCKD-EPI RIFLE score serum albumin mechanical ventilation (yes/no) CL cockcroft and gault serum creatinine site of infection comorbidity categories 'hematologic malignancy', 'oncologic malignancy', 'trauma' and 'head injury'		
Covariate inclusion method:	Univariate associations considered statistically significant at $p < 0.05$ following Likelihood Ratio Test.		
Covariates included*:	CKD-EPI Comorbidity hematologic malignancy Comorbidity trauma or head injury		
'Typical' patient for scaling:	CKD 73 mL/min		

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Graphical representation / schematic	NA
Equations	TV_CLCVVH = THETA_1 TV_CLnonCVVH = THETA_2* (CKD-EPI/median CKD-EPI population)^THETA_3 * THETA_4^(hemat¹) * THETA_5^(trauma/head injury²) TVV1= THETA_6 ¹hemat = 1 if comorbidity was hematologic malignancy, zero if otherwise. ²trauma/head injury = 1 if comorbidity was trauma or head injury, zero if otherwise.
Notes/further info:	

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	2.9			11	
THETA_2 (L/h)	3.42			9	
THETA_3	0.772			11	
THETA_4	1.57			17	
THETA_5	1.99			13	
THETA_6 (L)	46.8			12	
Between-subject variability (inter-individual variability)					
OMEGA_CL non CVVH (CV %)			36	14	
OMEGA_V1 (CV%)			102.8	18	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.281			12	
Notes/further info:					

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** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig 2	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig SA2	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

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Summary						
Model name:	Ceftaz_2		Population type*:		ICU	
Drug*:	Ceftazidime		Model type:		POPPK	
Administration route/mode:	IV/20 min		No. compartments*:		2	
Publication						
Title:	A population pharmacokinetic approach to ceftazidime use in burn patients: influence of glomerular filtration, gender and mechanical ventilation					
Author(s):	Jean Marie Conil, Bernard Georges, Michel Lavit, Jacky Laguerre, Kamram Samii, Georges Houin, Sylvie Saivin					
Other info:	PMID 17324245					
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/17324245/					
Source Study & Dataset						
Source study:	NA					
Inclusion criteria:	NA					
Patient disease(s):	Burn patients					
Drug dosing units*:	mg	Number of patients:		50		
Plasma conc. units*:	mg/L	No. samples:		237		
Daily dosing scheme	2000 q8h and 1000 q4h					
Notes/further info:						
Patient Characteristics (non obese/obsess/morbidly obese)						
Continuous covariate	Units	Median	Mean	SD	Mode	Range
Age	Y		52.3	20.7		15–90
Body weight	Kg		71.3	13.5		50–108
Burned surface area (% of the total body surface)	%		23.0	13.5		10–62
Baux index	%		75.6	22.4		30–122
UBS index			64.8	50.0		10–233
Tobiasen index	/16		7.0	1.8		4–11
Serum creatinine	mmol /L		76.5	21.8		35–141
Creatinine clearance	ml/min		105.3	39.3		33–191
Proteinaemia	g/L		56.8	7.1		38–69
BUN	mmol /L		7.3	3.6		2–19
Categorical covariate	Feature			Count (%)		
Sex Male	Yes			76		
Mechanical ventilation	Yes			68		
Notes/further info:						

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Final Model Structure			
Compartments:	2		
Elimination (e.g. 1 st order):	1 st		
Estimation algorithm:	NA	Log-transformed?	Yes
BLLOQ handling method (PIP) (mg/L)	1	Dose compartment:	Central
All covariates tested:	Age Baux index creatinine clearance (Cockcroft formula) serum creatinine Na pCO2 SaO2 alkaline phosphatases body weight proteinaemia BUN alkaline reserve gender transaminase burned area Tobiasen index mechanical ventilation		
Covariate inclusion method:	Final model : change in the objective function value of more than 10.83 (corresponding to a Chi-2 with 1 degree of freedom for a P value lower than 0.0001)		
Covariates included*:	CLCR, sex, ventilation status		
'Typical' patient for scaling:	NA		
Graphical representation / schematic	NA		
Equations	$TVCL = THETA_1 + THETA_2 * CLCR$ $TVV1 = THETA_3$ $TVQ = THETA_4$ $TVV2 = THETA_5 * (1 + THETA_6 * SEX) * (1 + THETA_7 * VENT) * (1 + THETA_8 * CLCR)$ with SEX = 0 for men and 1 for women, VENT = 1 if the burn patient was mechanically ventilated and = 0 if not		

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Notes/further info:	

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA1_ (L/h)	1.08				
THETA_2	0.0536				
THETA_3 (L)	18.8				
THETA_4 (L/h)	6.88				
THETA_5 (L)	2.69				
THETA_6	1.43				
THETA_7	2.44				
THETA_8	0.00414				
Between-subject variability (inter-individual variability)					
OMEGA_ CL (CV%)			16	Not estimated	
OMEGA_ V1			13	Not estimated	
OMEGA_ Q			Not estimated	Not estimated	
OMEGA_ V2			Not estimated	Not estimated	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.38				
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	Yes, fig 1
Other goodness-of-fit plots:	Yes, fig 2	Simulated plasma conc. profiles:	NA
Notes/further info:			

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Summary						
Model name:	Ceftaz_3		Population type*:		ICU	
Drug*:	Ceftazidime		Model type:		POPPK	
Administration route/mode:	IV/		No. compartments*:		2	
Publication						
Title:	Population Pharmacokinetics of Ceftazidime in Intensive Care Unit Patients: Influence of Glomerular Filtration Rate, Mechanical Ventilation, and Reason for Admission ▾					
Author(s):	Bernard Georges, Jean-Marie Conil, Thierry Seguin, Stéphanie Ruiz, Vincent Minville, Pierre Cougot, Jean-François Decun, Hélène Gonzalez Georges Houin Olivier Fourcade Sylvie Saivin					
Other info:	PMID 19635962					
Publication URL*:	https://pmc.ncbi.nlm.nih.gov/articles/PMC2764213/					
Source Study & Dataset						
Source study:	NA					
Inclusion criteria:	inpatient stay in the ICU over 18 years old presenting with <i>P. aeruginosa</i> nosocomial pneumonia or bacteremia with a strain thought to be sensitive to ceftazidime.					
Patient disease(s):	NA					
Drug dosing units*:	mg		Number of patients:		72	
Plasma conc. units*:	mg/L		No. samples:		443	
Daily dosing scheme	2000 over 30 min or 2000 continuously q8h after 2000 loading dose					
Notes/further info:						
Patient Characteristics						
Continuous covariate	Units	Median	Mean	SD	Mode	Range
Age	Y		58	17		
Body weight	Kg		76.8	15.8		
LBW (kg)	Kg		57.2	8.4		
Height (cm)	Cm		172	7		
SAPS I	/56		13.9	4.4		
SAPS II	/163		47.8	15.6		
Serum creatinine (μmol/liter)	μmol/liter		87	86.9		
Cockroft creatinine clearance (ml/min)	ml/min		123	65		
LBW creatinine clearance (ml/min)	ml/min		92	44		
Kirkpatrick creatinine clearance	ml/min		92	42		
MDRD	ml/min		121	55		
Proteins	g/liter		59	8		
BUN	mmol/liter		8.9	6.4		

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Hemoglobin	g/dl		9.7	2.2		
Leukocytes	G/mm ³		14.5	4.9		
Categorical covariate	Feature			Count (%)		
Sex Male	Yes			85		
Mechanical ventilation (no/yes)	Yes			83		
Admission for polytrauma	Yes			38		
Admission for postsurgical	Yes			26		
Admission for medical reason	Yes			36		
Notes/further info:						

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1 st order):	1 st		
Estimation algorithm:	FOCE	Log-transformed?	Yes
BLLOQ handling method (mg/L)	NA	Dose compartment:	Central
All covariates tested:	Age gender body weight (total body weight, lean body weight) height etiology of admission mechanical ventilation serum creatinine glomerular filtration rate (MDRD formula or Kirkpatrick formula) proteins blood urea nitrogen leukocyte counts haemoglobin C-reactive protein simplified acute physiology scores (SAPS)		
Covariate inclusion method:	Each covariate was examined in the structural model with a 0.05 level of significance for the likelihood ratio test. The resulting pharmacostatistical model was refined by independently deleting each covariate, with 0.001 as the level of significance for the likelihood ratio test.		
Covariates included*:	MDRD (mL/min), mechanical ventilation, etiology (polytraume, post-opérative, medical)		
'Typical' patient for scaling:	NA		
Graphical representation / schematic	NA		
Equations	$TVCL = THETA_1 + THETA_2 * MDRD$ $TVV1 = THETA_3 * (1 - MECH) + (THETA_4 * MECH) \text{ si } MECH = 1 \text{ et nonMECH} = 0$		

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	TVQ = THETA_5 TVV2 = THETA_6 ^ (POLY) * THETA_7 ^ (POSTOP) * THETA_8 ^ (MED) if POLY, POSTOP, MED = 1 or 0
Notes/further info:	49 patients for model building 23 for validation

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	2.24			27	
THETA_2	0.024			23	
THETA_3 (L)	18.90			10	
THETA_4 (L)	9.02			9	
THETA_5 (L/h)	15.20			13	
THETA_6 (L)	57.10			12	
THETA_7 (L)	25.70			18	
THETA_8 (L)	13.60			26	
Between-subject variability (inter-individual variability)					
OMEGA_CL	0.09			24	
OMEGA_V1	0.12			86	
OMEGA_Q	0.50			38	
OMEGA_V2	0.11			116	
Between-occasion variability (inter-occasion variability)					
KAPPA_CL	0.05			13	
Residual error					
Proportional error	0.2 (assumed, not reported in the original study)				
Notes/further info:	Prediction error (PE) available				

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Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	Yes, fig 1
Other goodness-of-fit plots:	NA	Simulated plasma conc. profiles:	Yes, fig 3
Notes/further info:			

Model scraping for comparison and validation

Summary						
Model name:	Ceftaz_4		Population type*:		ICU	
Drug*:	Ceftazidime		Model type:		POPPK	
Administration route/mode:	IV/continuous		No. compartments*:		1	
Publication						
Title:	Loading Dose of Ceftazidime Needs to Be Increased in Critically Ill Patients: A Retrospective Study to Evaluate Recommended Loading Dose with Pharmacokinetic Modelling					
Author(s):	Manon Launay, Edouard Ollier, Benjamin Kably, Félicien Le Louedec, Guillaume Thiery, Julien Lanoiselée, Sophie Perinel-Ragey					
Other info:	PMID 39200056					
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/39200056/					
Source Study & Dataset						
Source study:	NA					
Inclusion criteria:	NA					
	Exclusion: renal replacement therapy					
Patient disease(s):	NA					
Drug dosing units*:	mg	Number of patients:		86		
Plasma conc. units*:	mg/L	No. samples:		223		
Daily dosing scheme	NA					
Notes/further info:	External validation performed					
Patient Characteristics						
Continuous covariate	Units	Median	Mean	SD	Mode	Range
Age	Y		64.5	11.9		
Weight	Kg		91.3	25.3		
BMI	kg/m²		31.2	9.5		
GFR	mL/min/1.73 m²		87.6	74.2		
Categorical covariate	Feature			Count (%)		
Sex male	Yes			78		
Notes/further info:						

Final Model Structure			
Compartments:	1		
Elimination (e.g. 1 st order):	1 st		
Estimation algorithm:	NA	Log-transformed?	Yes

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BLLOQ handling method (PIP) (mg/L)	4	Dose compartment:	Central
All covariates tested:	Age sex weight height status of ongoing COVID infection BMI serum creatinine at baseline and at ceftazidime sampling protein concentration ceftazidime regimen and serum concentrations eGFR by CKD-EPI formula		
Covariate inclusion method:	Stepwise covariate modelling significance level for covariate incorporation alpha = 0.05.		
Covariates included*:	eGFR mL/min/1.73 m ²		
'Typical' patient for scaling:	73.9		
Graphical representation / schematic	NA		
Equations	$TVCL = THETA_1 * (GFRi / GFRmedian)^{THETA_2}$ $TVV1 = THETA_3$		
Notes/further info:			

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h)	4.45			7.3	
THETA_2	0.9			15.5	
THETA_3 (L)	88			18.3	
Between-subject variability (inter-individual variability)					
OMEGA_CL	0.46			11.5	38.9
OMEGA_V1	0.57			25.6	82.5
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
additive error (mg/L)	0.39			6.2	
Notes/further info:					

** required as a minimum for model replication*

Model Evaluation Metrics			
Does the model publication provide the following?			
Visual predictive check (VPC) plot(s):	Yes, fig S1	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	Yes, fig 1	Simulated plasma conc. profiles:	Yes, fig 2, S3
Notes/further info:			

Model scraping for comparison and validation

Summary						
Model name:	Cefta_5	Population type*:	ICU			
Drug*:	ceftazidime	Model type:	POPPK			
Administration route*:	IV/30 min	No. compartments*:	2			
Publication						
Title:	Population pharmacokinetics of four β -lactams in critically ill septic patients comedicated with amikacin					
Author(s):	Isabelle K. Delattre, Flora T. Musuamba , Philippe Jacqmin, Fabio S. Taccone, Pierre-François Laterre, Roger K. Verbeeck, Frédérique Jacobs , Pierre Wallemacq					
Other info:	PMID 22503878					
Publication URL*:	https://pubmed.ncbi.nlm.nih.gov/22503878/					
Source Study & Dataset						
Source study:	NA					
Inclusion criteria:	patients with severe sepsis or septic shock treated with a loading dose of AMK (25 mg/kg) combined with ceftazidime					
Patient disease(s):	sepsis or septic shock					
Drug dosing units*:	mg	Number of patients:	18			
Plasma conc. units*:	mg/L	No. samples:	NA			
Daily dosing scheme	2000 q8h					
Patient Characteristics (adjust as appropriate)						
Continuous covariate	Units	Median	Mean	SD	Mode	IQR
NA						
Categorical covariate	Feature			Count (%)		
NA						
Notes/further info:	PKPOP model for meropenem, piperacillin, ceftazidime, cefepime					

Final Model Structure			
Compartments:	2		
Elimination (e.g. 1 st order):	1 st		
Estimation algorithm:	FOCEI	Log-transformed?	Yes
BLLOQ handling method (mg/L):	NA	Dose compartment:	Central
All covariates tested:	Renal function		
Covariate inclusion method:	Empirical selection		
Covariates included*:	Creatinine clearance (Cockcroft–Gault), body weight		

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'Typical' patient for scaling:	CLCrstd 100 mL/min, BWstd 70 Kg
Graphical representation / schematic	NA
Equations	$TVCL = THETA_1 * (CLC_{ri}/CLC_{rstd})^{THETA_2} * (BW/BW_{std})^{0.75}$ $TVV1 = THETA_3 * (BW/BW_{std})^1$ $TVQ = THETA_4 * (BW/BW_{std})^{0.75}$ $TVV2 = THETA_5 * (BW/BW_{std})^1$
Notes/further info:	Model 1

Parameter Estimates* (adjust as appropriate)					
PK parameter (units)	Value	95% CI	CV%	RSE	Shrinkage
Fixed effects					
THETA_1 (L/h/70kg)	3.91			17.9	
THETA_2 (power)	0.29			62.6	
THETA_3 (L/70kg)	20.3			15.4	
THETA_4 (L/h/70kg)	17.9			18.4	
THETA_5 (L/70kg)	14.9			15.5	
Between-subject variability (inter-individual variability)					
OMEGA_CL (CV %)			58.5	26.9	
OMEGA_V1 (CV %)			63	35.3	
OMEGA_Q			Not estimated	Not estimated	
OMEGA_V2 (CV %)			42.2	28.4	
Between-occasion variability (inter-occasion variability)					
NA					
Residual error					
Proportional error	0.218			19.1	
Notes/further info:					
Model Evaluation Metrics					
Does the model publication provide the following?					

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Visual predictive check (VPC) plot(s):	NA	Example plasma conc. profiles:	NA
Other goodness-of-fit plots:	NA	Simulated plasma conc. profiles:	NA
Notes/further info:			